Development Operations

Mini Project Assignment # 3

REG#: ________________ NAME: ________________________________

COURSE CODE: CS316 INSTRUCTOR: MUHAMMAD SAJID ALI

TOTAL MARKS: 100

Automating DOMjudge Deployment Using Bash Script and Docker Compose

Objective

The objective of this assignment is to automate the deployment and management of the DOMjudge system, an open-source platform for programming contests using Bash scripting and later with Docker Compose. Students will learn how to configure, deploy, and manage multiple interconnected services (DOMserver, Judgehost and database) through automation, reinforcing their understanding of container orchestration, service networking, and environment configuration.

General Instructions:

DOMjudge is an open-source system for running programming contests, comprising multiple components:

- domserver – the central web server and API backend.
- judgehost – the worker node that fetches submissions from the domserver and evaluates them.

For this assignment, you will automate the deployment of DOMjudge using official Docker images:

- domjudge/domserver:8.2.0 (https://hub.docker.com/r/domjudge/domserver)
- domjudge/judgehost:8.2.0 (https://hub.docker.com/r/domjudge/judgehost).

After deployment, verify that the judgehosts are successfully connected to the DOMjudge server.

- To do this, log in to the DOMjudge web interface as an administrator and navigate to the "Judgehosts" menu. You should see the registered judgehost containers listed there, along with their status.

Deliverables & Submission:

You should submit the following structure in a zip named as `A2_<your-reg-number-with-u>.zip`:

- Task_1/
    - domjudge-deployment.sh
    - README.md

- Task_2/
    - docker-compose.yml
    - Any script used in docker compose.
    - README.md

Task 1: Apply Bash scripting to automate deployment of the DOMjudge system. (CLO2, GA-5, C3) **(60 marks)**

Create a Bash script named `domjudge-deployment.sh` in the current directory that automates the deployment and management of the DOMjudge system.

Use the official documentation and Docker images for configuration:

- Docker Hub: https://hub.docker.com/r/domjudge/domserver
- Official Documentation: https://www.domjudge.org/docs/latest/install.html

Functional Requirements:

- Your script should control all operations via command-line arguments and manage containers, networks, and volumes automatically.

Argument	**Purpose**	**Behavior**
`-i`	Install / Deploy	Deploys app services (domserver, judgehost, and database). If no argument is provided, assume -i by default.
`-dt`	Delete	Deletes all services and configurations but preserves volumes.
`-sp`	Stop	Temporarily stops all running services.
`-st`	Start	Starts all previously stopped services.

- Accept an optional argument (e.g., `--judgehosts=3`) to create multiple judgehost containers. Default is one.
- Create a Docker network prefixed by the current directory name (e.g., `mydir_domjudge_net`).
- Prefix all container and volume names with the current directory name (i.e. `mydir_server`, `mydir_judge1`).
- Create a persistent volume for the database, also prefixed by the directory name.
- After deployment, list all running containers, networks, volumes, generated secrets, and any manual steps required.

Example Commands:

- `./domjudge-deployment.sh`
- `./domjudge-deployment.sh -i`
- `./domjudge-deployment.sh -i --judgehosts=3`
- `./domjudge-deployment.sh -sp`
- `./domjudge-deployment.sh -st`
- `./domjudge-deployment.sh -dt`

Task 2: Apply Docker Compose to convert the Bash-based deployment into a declarative multi-container setup. (CLO2, GA-5, C3) **(40 marks)**

Convert your Bash-based automation into a Docker Compose setup. Create a docker-compose.yml file that replicates the same behavior as your script. Service names, networks, and volumes will be managed automatically by Docker Compose.